The categorification frontier

Six open cases, branch exclusions and the pivotal parity obstruction

Six rings remain unresolved. The general-complex exclusion of `R8_3_1768` is proved in Appendix A. The six open cases have no newly established categorification or whole-ring exclusion. Each case was examined separately, with common modular, Galois, indicator and Morita arguments used where possible. The strongest new exclusions concern necessary branches of two cases, rather than all possible categorifications.

Case	Verified progress
`R6_4_155`	Within the specified order-32 near-group route, the parent group must be $C_4 \times C_2^3$. Its existence remains open.
`R6_8_1660`	A nonreal Morita invariant rules out real forms and braided representatives in the positive Morita class; smaller direct modular sources are excluded.
`R8_3_928`	Three center patterns are excluded. The remaining rank-20 pattern now has exact candidate modular data and a simple fusion ring. See pages 4–6.
`R8_3_941`	The center is prime; a uniform direct-source obstruction applies. The proposed Morita link with R1660 is still conditional.
`R8_3_1154`	A smallest possible algebra has two based-module branches. One would force an explicit simple rank-7 Morita-dual ring.
`R9_2_137`	The positive Tambara–Yamagami sign is excluded by nine exact certificates. The negative sign survives stronger necessary conditions.

Meaning of the status. A candidate fusion ring, a based module, modular matrices, or a rational solution of necessary equations is not a categorification. A construction needs the required category and algebra structures, including associators. No such new structure is asserted here.

Mathematical scope. Center arguments are applied to the positive spherical representative supplied by the project's previously established general-complex existence reduction. The complete center domains and certain indicator reductions are inherited premises, explicitly identified below. We do not assume that every original complex categorification is unitary or already positive. Unitarity of the normalized modular *matrix* is not an assumption of unitary associators.

Source and verification. The six input tensors were recovered from the classification repository at commit

`536be977f50b98fcc21d2640b7798cbac3ec1c20`.

They are pinned by tensor hashes. New arguments received separate agent reviews and exact finite replays; this is not external human verification. The companion proof directory includes current proofs, exact inputs and optimizer-free checks. Current proofs, source bindings and exact replay are maintained in the classification repository.

1. A common obstruction from twist-one rows

Let S be a symmetric unitary normalized modular matrix, T the actual twists, and ξ the anomaly, so

$$STS = \xi T^{-1} S T^{-1}, \qquad |\xi| = 1.$$

Let J consist of twist-one labels. For $X \notin J$, assume S_{Xj} is real for $j \in J$, and put

$$m_X = \sum_{j \in J} S_{Xj}^2.$$

The diagonal modular relation and the row norm give

$$m_X \le 1 - m_X + |S_{XX}| \le 1 - m_X + \sqrt{1 - m_X}, \qquad \boxed{m_X \le \tfrac{3}{4}}.$$

Indeed the known contribution to $(STS)_{XX}$ is m_X, while the remaining terms have total absolute value at most $1 - m_X$.

Equality. If J is duality-closed and $m_X = 3/4$, equality forces every outside-J entry other than S_{XX} to vanish and $|S_{XX}| = 1/2$. If $X \neq X^*$, the two conjugate rows have equal real restrictions to J, of squared norm $3/4$, and cannot be orthogonal. Thus $X = X^*$ and $S_{XX} = \pm 1/2$. Now

$$\tfrac{3}{4} + \tfrac{1}{4}\theta_X = \xi \theta_X^{-2} S_{XX}$$

forces $\theta_X = -1$, $\xi = \pm 1$, and $S_{XX} = \xi/2$. For the Drinfeld center of a spherical fusion category, with its canonical normalization, $\xi = 1$.

Strict inequality for `R9_2_137`

Here J is the set of nine distinguished constituents of $I(1)$. For a restriction vector $v = [F(X)]$ outside J, $v_0 = 0$. Write

$$a = v_1 + v_2 + v_7 + 2v_8, \quad b = v_3 + v_4 + v_5 + v_6, \quad d = 4 + 3\sqrt{2}, \quad D = 144 + 96\sqrt{2}.$$

The known rows are $S_{A_E,X} = \chi_E(v)/f_E$. Equality in the mass bound would imply, by orthogonality with the unit,

$$\sum_E \frac{\chi_E(v)}{f_E^2} + \frac{a + bd}{2D} = 0.$$

The exact characters reduce this to $b = 2a$ and $v_2 + v_7 = 3v_1 + 4v_8$. With $p = v_1$, $q = v_8$, the mass then becomes

$$\frac{9p^2 + 24pq + 21q^2 + (v_2 - v_7)^2 + (v_3 - v_6)^2 + (v_4 - v_5)^2}{16} + \frac{(v_3 + v_6 - v_4 - v_5)^2}{64}.$$

For this to equal $3/4$, nonnegativity forces $p = 1, q = 0$ and $(v_2 - v_7)^2 = 1$. Hence $b = 8$ and

$$(v_3 - v_6)^2 + (v_4 - v_5)^2 + \tfrac{1}{4}(v_3 + v_6 - v_4 - v_5)^2 = 2.$$

If $v_3 + v_6$ and $v_4 + v_5$ are even, the left side is 0 modulo 4; if odd, it is 3 modulo 4. Both contradict 2. Therefore **every noncanonical integral restriction column has** $m_X < 3/4$. This proof has no bound on the size of v.

2. R9_2_137: one TY sign is impossible

The ring has four invertibles, one object of dimension 2, and four objects $X_3, \dots, X_6$ of dimension $d = 4 + 3\sqrt{2}$. Its rank-5 subcategory has Tambara–Yamagami (TY) type. The two possible TY signs are treated separately.

Full-column Galois complement. For $v = [F(Z)]$, put $a = v_0 + v_1 + v_2 + v_7 + 2v_8$. The signed Galois rule for the nine known character rows gives an exact involution

$$(Tv)_i = v_i \quad (i = 0, 1, 2, 7, 8), \qquad (Tv)_i = a - v_i \quad (i = 3, 4, 5, 6).$$

For $a > 0$, positivity fixes the sign to $+1$, so each large coordinate satisfies $v_i \le a$. For $a = 0$, the sign is negative and the physical restriction vector is unchanged. For $a > 0$, total multiplicities of $(v, \text{joint twist-trace type})$ and $(Tv, \text{same type})$ must agree. Duplicate variables are aggregated. Deleting types with no partner uses the inherited *complete* domain theorem.

An integral condition absent from the old relaxation. Let B be the Galois conjugate of the central unit. The known rows force

$$F(B) = 1 + X_3 + X_4 + X_5 + X_6, \qquad S_{B,Z} = \frac{a + (4a - b)d}{D}, \quad b = \sum_{i=3}^{6} v_i.$$

Verlinde makes all nine $n_E = N_{BB}^{A_E}$ linear functions of central-simple multiplicities. Each must be a nonnegative integer, with vanishing irrational part. For example,

$$N_{BB}^{B} = \sum_{Z} \frac{(a + (4a - b)d)^3}{D^2(a + bd)}.$$

The old positive-sign rational witness gives $N_{BB}^{B} = 5/6$; this rejects that witness, by itself not the whole sign.

Uniform contradiction for the positive sign. The inherited positive-TY proof forces all four projected third indicators to 1/2. The four old test equations setting fourth indicators to 2 are *removed*. Nine exact rational certificates, checked on all 5,996 allowed columns, give successively:

Coefficients	Certified bound	Integral consequence
n_B	$1/2 \le n_B \le 31/18$	$n_B = 1$
Two codegree-16 coefficients	$0 \le n_E \le 17/30$	Both 0
Four codegree-8 coefficients	$0 \le n_E \le 3/4$	All 0
Codegree-4 coefficient	$0 \le n_E \le 1$	At most 1

But adjunction gives $\sum_E n_E = [F(B)^2 : 1] = 5$. The unit contributes 1, B contributes 1, and all other contributions total at most 1: contradiction. **The positive TY sign is excluded.**

What survives. The negative sign has an independently checked exact rational witness with 124 positive weights, all 45 tested distinguished fusion coefficients integral and nonnegative, and the strict mass bound from page 2. It has projected third indicators 1/2 and fourth indicators 2; these are witness choices, not a theorem forcing them in every negative-sign category. This remains a necessary-system witness, not modular data or a model.

3. R8_3_928: modular patterns and twists

The inherited reduction gives

$$Z(\mathcal{C}) = \mathrm{Sem} \boxtimes \mathcal{B}, \quad D = 144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6}, \quad L = D/\sqrt{2},$$

where $\mathcal{B}$ is simple, has global dimension L^2, and has four possible dimension patterns numbered $40, 49, 186, 191$, of ranks $23, 20, 20, 17$. Fix anomaly $\xi(\mathcal{B}) = \zeta_8$; conjugation covers the other orientation.

Patterns 49 and 191 are impossible. The four scalar-character constituents of $I(1)$ belong directly to $\mathcal{B}$: their semion charge is positive and their twists are 1. A required object H of dimension $L/2$ has normalized entries

$$(S_{H,A_0}, S_{H,A_1}, S_{H,A_2}, S_{H,A_3}) = (1, -1, -1, 1)/2.$$

These exhaust its row norm. Thus all other entries, including S_{HH}, vanish. The diagonal modular relation gives $(STS)_{HH} = 1$ and also $(STS)_{HH} = 0$, a contradiction independent of every unknown twist.

Finite twist domains are justified, not guessed. In the retained patterns, the full square-Galois orbits preserve each repeated dimension fiber. The two singleton dimensions E, F form one orbit and their twists are equal by a known cross-STS entry. Cauchy's theorem and $\mathrm{Norm}(D) = 2^{14}3^2$ restrict prime support to $2, 3$. A root of order 2^a3^b has square-orbit size

$$2^{\max(a-3,0)}3^{\max(b-1,0)}.$$

Fiber multiplicities therefore bound the orders. Exact root-sum, Vafa and indicator equations then leave two twist allocations per pattern before the following block argument.

Integral projections become matchings. The remaining part Q of a known fusion matrix is symmetric integral, positive semidefinite, of rank 3, and satisfies $Q^2 = 2Q$ and $Q\mathbf{d} = 0$. A diagonal 2 would force an isolated nonzero diagonal, contrary to $Q\mathbf{d} = 0$. A diagonal 1 has exactly one off-diagonal entry, necessarily -1, on an equal-dimension label. Thus

$$Q = \text{three disjoint pair blocks } \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

This replaces an unknown real projection by a short exact matching problem.

A six-dimensional block. The four H twists have the form $a(1, i, -1, -i)$, with two antipodal matching pairs. The third pair has common twist λ. Known row projections force a six-dimensional invariant S, T block. In pair-sum and pair-difference coordinates, its unknown real symmetric involution is

$$M = \xi^{-1} A R \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} R^t A, \quad A = \mathrm{diag}(a, ia), \quad R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The condition $M^2 = I$ forces $a^4\lambda = \xi^2 = i$. Only $a = \zeta_{32}^3$ and the pair of twist $\lambda = \zeta_8^7$ survive. No identification of this block with a known categorical construction is asserted.

4. R8_3_928: a third pattern is excluded

Pattern 40 has rank 23. Normalize the linear modular representation by $t = T/\zeta_{24}$. The six-dimensional block from page 4 and the known two-dimensional block on the canonical sum and W are invariant. Removing them leaves a 15-dimensional representation with t-spectrum

exponent modulo 24	3	9	11	12	15	17	20	23
multiplicity	2	2	1	4	2	1	1	2.

The full representation factors through $SL_2(\mathbb{Z}/96)$ by the modular congruence theorem. The complement in fact factors through level 24: the kernel of $SL_2(\mathbb{Z}/32) \to SL_2(\mathbb{Z}/8)$ is the additive group

$$\{I + 8M : \ M \in \mathrm{Mat}_2(\mathbb{Z}/4),\ \mathrm{tr}\, M = 0\},$$

of order 64, and is generated normally by t^{24}, which acts trivially on the complement. Conjugates give the two off-diagonal directions and the diagonal trace-zero direction.

Three sign characters are forced. The complete character tables of $SL_2(\mathbb{Z}/8)$ and $SL_2(\mathbb{Z}/3)$ give all 210 irreducibles of level 24. Only 12 fit the displayed spectrum. Their multiplicity equations force three copies of the one-dimensional character $S = t = -1$. This can be checked by two linear identities on their spectral multiplicities; it does not need a search for associators. Both central signs are included, so no unproved self-duality assumption enters.

Signed Galois symmetry forbids them. Let $G_5 = \sigma_5(S)S^{-1}$. The Galois word

$$G_5 = t^5 S t^{77} S t^5 S^{-1}$$

acts as $+1$ on each sign character. It therefore has at least three fixed vectors in the full $t = -1$ eigenspace. That eigenspace consists of exactly four equal-dimension simple-object coordinates, each with $S_{0A} = 1/(4\sqrt{3})$. Since $\sigma_5(\sqrt{3}) = -\sqrt{3}$, signed Galois symmetry gives $G_5 = -P$ on this space for a four-point permutation P. Its fixed-space dimension is the number of even cycles of P, at most two. This contradiction excludes pattern 40.

More generally, a Galois-stable m-object twist fiber with all negative Galois signs has at most $\lfloor m/2 \rfloor$ fixed vectors. Any forced representation contributing more fixed vectors in that fiber excludes the candidate.

Only pattern 186 remains. It has rank 20. With $\zeta_n = e^{2\pi i/n}$, its unique retained twist multiset, up to relabeling and conjugation, is

fiber	twists
four canonical labels	$1, 1, 1, 1$
W	i
E, F	ζ_3^2, ζ_3^2
A	$\zeta_6, \zeta_6, -1, \zeta_8^7, \zeta_8^7$
B	$\zeta_{12}^5, \zeta_{12}^5, -i$
K	ζ_{24}^{13}
H	$\zeta_{32}^3(1, i, -1, -i)$.

The matching pairs are the antipodal H pairs and the two A objects of twist ζ_8^7.

An exact complete candidate. Explicit 20×20 matrices S, T and their simple, self-dual fusion ring are now supplied in the archive. Separate exact replays verify $S^2 = I$, STS, all 8,000 Verlinde identities, associativity, balancing and indicator constraints; the full Vafa equations pass as well. The candidate has $\nu_2 = 1$ on every simple. This is a modular-data construction, not a fusion category or an algebra construction; uniqueness of the full data is not asserted. R928 remains unresolved.

4a. Identification in The Anyon Atlas

The explicit rank-twenty modular datum is the existing record B20-02148 of *The Anyon Atlas*, Galois orbit 20.167. It is the proposed simple factor $\mathcal{B}$ of $Z(\mathcal{C}) = \mathrm{Sem} \boxtimes \mathcal{B}$, not the whole center. Its global dimension is

$$38592 + 27264\sqrt{2} + 22272\sqrt{3} + 15744\sqrt{6} = \tfrac{1}{2}(144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6})^2.$$

It has central charge $1 \bmod 8$, twist order 96, and a simple, self-dual fusion ring of maximum multiplicity 18. The semion factor has the opposite central charge $-1 \bmod 8$; their product has central charge $0 \bmod 8$, as the Drinfeld center of a spherical fusion category should.

This is an exact identification of both modular matrices and the full based ring, not merely of their numerical invariants. With zero-based project labels, the relabeling is

$$(p_0, \ldots, p_9) = (0, 7, 11, 18, 19, 1, 2, 3, 4, 5),$$
$$(p_{10}, \ldots, p_{19}) = (6, 8, 9, 10, 12, 13, 14, 15, 16, 17).$$

Writing $L = D/\sqrt{2}$, the Atlas normalization has $S_{00} = 1$ and satisfies $S^{\mathrm{Atlas}}_{p_i p_j} = L S^{\mathrm{project}}_{ij}$, $T^{\mathrm{Atlas}}_{p_i} = T^{\mathrm{project}}_{i}$. A safe parser converts the downloaded cyclotomic expressions to $\mathbb{Q}(\zeta_{288})$. All 400 matrix entries, 20 twists and 8000 structure constants agree exactly. The original export, comparison program and hash-bound receipt are in the current proof directory's `r928/atlas/`.

As accessed on 29 October 2026, the record is labeled *Not CS* and is not selected by *Known Categorifiable*. Its proposed condensation of $1 + X_7 + X_{11} + X_{18}$ to the semion record B02-00007 has algebra dimension $D/2$, but is not guaranteed by the export. Neither this possible condensation nor the exact modular-data match constructs the category or its algebra. No novelty of the modular datum, and no categorification of R928, is asserted.

5. R8_3_1154: a conditional rank-7 route

Put $a = 6 + 3\sqrt{5}$, $b = a + 1$, $\delta = (5 + 3\sqrt{5})/2$, and $D = 540 + 240\sqrt{5}$. The pointed C_2 subcategory has nontrivial associator. Its nontrivial invertible g therefore acts freely on every module category: a fixed simple would give a rank-one module for this nontrivially associated pointed category.

Consequently no connected separable algebra contains g. The smallest *possible* nontrivial algebra has object $1 + X_4$ or $1 + X_5$, of dimension b. Existence is not implied.

Two six-vertex branches. Assuming an actual algebra $1 + X_4$, its free-induction Gram matrix forces exactly six simple modules, of relative dimensions

$$(1, 1, \delta, \delta, 2, 2).$$

The induction matrix has rank five. Its missing line can carry exactly two source characters, χ_4 or χ_{20}. Both yield nonnegative integral module matrices satisfying all source products and duality. This is only a based-module calculation.

The χ_4 branch forces a dual ring. The module-character multiplicities are $(1, 1, 2, 0, 1)$ on the five source irreducible representations. Ostrik's linking-multifusion character identification transfers these to the distinguished constituents of the module's Lagrangian algebra. Their weighted dimension is already

$$1 + (161 + 72\sqrt{5}) + 2(135 + 60\sqrt{5}) + (108 + 48\sqrt{5}) = D.$$

Positivity leaves no other summands. Its squared-multiplicity sum is 7; hence the Morita dual has rank 7 and acts faithfully as the full commutant of the six-vertex module.

For a nonunit simple commuting matrix, the coefficient-of-unit trace is zero and its squared trace norm is one. The dimension bound makes the resulting integral parameter domain small. Exactly two orthonormal six-element choices survive; one has nonintegral or negative products. The other is the unique possible nonnegative integral based ring, with dimensions

$$\boxed{(1, \delta, \delta, \delta + 1, \delta + 1, 2\delta + 1, 4\delta + 4)}$$

and duality $(0, 1, 2, 4, 3, 5, 6)$. It is simple and noncommutative, with maximum multiplicity 19. Its full tensor is supplied in the proof directory.

Morita-invariant total second and third indicators force

$$\nu_2 = (1, 1, 1, 0, 0, 1, 1), \qquad \nu_3 = (1, 1, 1, 1, 1, 2, 4).$$

These are consistent. The complete zero- and one-spectrum tests also give no obstruction, for either this candidate or the original R1154 tensor.

Exact remaining gap. No algebra structure on $1 + X_4$ or $1 + X_5$ is constructed. The χ_{20} branch also remains possible. Thus the rank-7 ring is a *conditional* Morita partner, not a categorification of R1154 and not an unconditional reduction of every possible R1154 category.

6. R6_4_155: a restricted near-group parent

This is the non-self-dual ring $R'(4)$, with $X^* = Y$ and

$$XY = 1 + h + 4(X + Y), \qquad X^2 = g + hg + 4(X + Y).$$

Meng–Yu's August 2026 paper explicitly leaves this case open. Its construction gives $R'(2)$, not $R'(4)$. The inherited rank-26 necessary modular data do not supply associators.

A conditional parent theorem. Consider a proposed order-32 near-group route with the following precise low-center hypotheses: its invertibles form an abelian group G of order 32 with nondegenerate symmetric bicharacter χ and twists $q(g) = \chi(g, g)^{-1}$; all noninvertible central simples have dimension greater than 8; and an étale algebra E of dimension 8 has local modules equal to the proposed R155 center.

Then E is pointed, supported on an isotropic subgroup H of order 8. The monodromy is $b(g, h) = \chi(g, h)^{-2}$ and its radical is $G[2]$. The four target invertibles imply $H^{\perp_b} = G$, hence

$$2G \subset H \subset G[2], \qquad q|_H = 1.$$

The only groups to consider are $C_4^2 \times C_2$, $C_4 \times C_2^3$ and C_2^5. The last has no semion twists. In the first, $H = G[2]$; nondegeneracy of the order-four pairing and twist-freeness of the order-two generator force a radical element, a contradiction. Therefore

$$\boxed{G = C_4 \times C_2^3.}$$

The remaining metric data do exist. For $x = (a, b_1, b_2, b_3)$, set

$$\chi(x, x') = i^{aa'}(-1)^{b_1b_1' + b_2b_2' + b_3b_3'}, \qquad H = \{a \text{ even},\ b_1 + b_2 + b_3 \text{ even}\}.$$

The quotient has twists $1, -1, -i, i$. This verifies compatibility of the pointed data, not existence of the near-group category or the required algebra.

Radchenko–Wheeler's September 2026 construction supplies groups that are products of two cyclic groups; its existence corollary does not cover this four-generator group. Its broader reconstruction theorem still needs an additional finite-quantum-dilogarithm input. No such input is provided here.

Scope of the result. The parent theorem applies only under the explicitly listed low-center hypotheses. It is not a classification of arbitrary spherical near-group categories. In particular no complete categorical parameterization is inferred from a unitary table. R155 remains unresolved.

A limitation shared by all six cases. Longer palindromic conjugators and direct sums do not strengthen the already imposed pivotal conjugation-parity graph: the new cross terms form hyperbolic pairs. Exact tensor checks confirm that the existing sign assignments for all six cases pass these saturated parity restrictions. Repeating that parity test cannot resolve the remaining cases.

7. R6_8_1660 and R8_3_941

Both rings have global dimension

$$D = 144 + 96\sqrt{2} + 80\sqrt{3} + 56\sqrt{6}.$$

This equality motivates a Morita search; it does not establish a Morita equivalence. R928 also has this global dimension and illustrates the limitation of dimension matching.

A nonreal Morita invariant for R1660. Put $d = 3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$. The inherited complete third-indicator theorem gives, in the positive representative,

$$W_3(\mathcal{C}) := \sum_{V \in \operatorname{Irr}\mathcal{C}} d_V \nu_3(V) = 1 + (3 + 2\sqrt{2})d \ \pm \ i\sqrt{3}d.$$

This is nonreal. For a positive category, the center indicator formula gives

$$W_n(\mathcal{C}) = \frac{1}{\operatorname{FPdim}(\mathcal{C})} \sum_{Z \in \operatorname{Irr} Z(\mathcal{C})} d_Z^2 \theta_Z^n.$$

It is preserved by Morita equivalence, while complex conjugation and tensor opposition conjugate its value. Thus this positive Morita class is not equivalent to its conjugate or reverse, admits no real form, and contains no braided category. A braiding would supply an equivalence to tensor opposition.

This does not exclude unitarity: unitary associators need not admit a real gauge. It also does not transfer automatically to every original nonpositive category. For R941 it applies only if the proposed Morita equivalence is actually constructed.

A uniform direct-source obstruction for both rings. In both positive centers, the inherited integral part is sVec and every nonpointed subcategory has dimension greater than D. If a proper nontrivial nondegenerate subcategory existed, it and its centralizer would both have dimension greater than D, contradicting the product formula D^2. Thus each center is prime.

Suppose a modular category $\mathcal{P}$ and a connected étale algebra A directly realize $\mathcal{C} = \mathcal{P}_A$. The standard center decomposition gives

$$Z(\mathcal{P}_A) \simeq \mathcal{P} \boxtimes (\mathcal{P}_A^0)^{\mathrm{rev}}.$$

Primeness forces $\mathcal{P}_A^0 = \mathrm{Vec}$: A must be Lagrangian and $\mathcal{P} \simeq Z(\mathcal{C})$. In particular **every smaller direct modular source is ruled out at once**. Noncentral and other indirect constructions remain possible.

A concrete coefficient obstruction. For $\mathcal{P} = SU(2)_6 \boxtimes SU(2)_{10}$, the required algebra dimension is

$$\frac{\operatorname{FPdim}(\mathcal{P})}{D} = 24 - 8\sqrt{6}.$$

All source object dimensions have nonnegative coordinates in the independent basis

$$1, \sqrt{2}, \sqrt{3}, \sqrt{6}, c, c\sqrt{2}, c\sqrt{3}, c\sqrt{6}, \qquad c = \sqrt{2 + \sqrt{2}}.$$

The negative $\sqrt{6}$ coefficient makes this impossible. The same argument excludes the doubled-source local construction of either center. These are route exclusions, not exclusions of the two rings.

8. Proof dependencies and reproduction

The companion proof directory separates new arguments, frozen inherited inputs and exact verification. Its `README.md` gives the entry point and required packages; the runner verifies hashes and executes the checks in a temporary copy. No numerical optimizer is needed for replay. Any finite searches are over proved finite domains, not open-ended associator searches.

Conclusion	Essential mathematical premises beyond tensor arithmetic
R137 positive-sign exclusion	General-complex existence reduction; complete center-column domain; previously proved positive-TY projected cubic values; modular Galois and Verlinde formulas.
R928 pattern reductions	Positive modular-factor reduction; complete four-pattern dimension list; identification and twists of the distinguished central rows; congruence and Galois theorems where used.
R1154 rank-7 route	An actual connected separable algebra in the specified module branch; positive full-center/module-character correspondence. The algebra's existence is not established.
R155 parent restriction	The three explicitly stated low-center hypotheses. They are not asserted for every unconstructed parent.
R1660/R941 routes	The inherited center-subcategory restrictions; R1660's complete third-indicator theorem for its nonreal invariant. The Morita link remains unproved.

Limits of verification. Rational separators prove incompatibility of their encoded necessary system; the categorical theorem making that system complete is a separate premise. A surviving rational witness proves only that those necessary linear tests do not yet contradict one another. Agent reviews are additional checks, not a claim of independent human validation.

Selected primary references. Detailed per-case notes give theorem numbers and additional sources.

[1] V. Ostrik, *Pivotal fusion categories of rank 3*, Remark 2.19 and Theorem 2.13. https://arxiv.org/abs/1309.4822.

[2] C. Dong, X. Lin and S.-H. Ng, *Congruence property in conformal field theory*, modular Galois and congruence theorems. https://arxiv.org/abs/1201.6644.

[3] T. Deshpande, *On centers of bimodule categories and induction-restriction functors*. https://arxiv.org/abs/1611.04486.

[4] S.-H. Ng and P. Schauenburg, *Frobenius–Schur indicators and exponents of spherical categories*. https://arxiv.org/abs/math/0601012.

[5] A. Davydov, M. Müger, D. Nikshych and V. Ostrik, *The Witt group of non-degenerate braided fusion categories*. https://arxiv.org/abs/1009.2117.

[6] Y. Meng and Z. Yu, *Classification of some $\mathbb{Z}/2 \times \mathbb{Z}/2$-quadratic fusion categories of rank 6*. https://arxiv.org/abs/2608.06064.

[7] D. Radchenko and C. Wheeler, *Real quadratic fields and finite quantum dilogarithms I*, Section 5.4. https://arxiv.org/abs/2609.21892.

[8] *The Anyon Atlas*, record B20-02148, https://anyonatlas.org/data/B20-02148, accessed 29 October 2026.

Appendix A. Pivotal parity and R1768

Lemma. *Let $\mathcal{C}$ be a pivotal semisimple rigid $\mathbb{C}$-linear category. Suppose V has a nondegenerate self-duality $f : V \to V^*$ of sign $\varepsilon \in \{1, -1\}$, namely*

$$f^* j_V = \varepsilon f.$$

For every X, the object $X \otimes V \otimes X^*$ has a nondegenerate self-duality of sign ε. Consequently, every self-dual simple S with second indicator $\nu_2(S) = -\varepsilon$ occurs in it with even multiplicity.

Proof. Suppress the canonical tensor-dual comparison maps and associators. Put $W = X \otimes V \otimes X^*$ and

$$h = j_X \otimes f \otimes \mathrm{id}_{X^*} : W \longrightarrow W^*.$$

Using monoidality of the pivotal structure and $j_{X^*} = (j_X^*)^{-1}$, one obtains

$$h^* j_W = j_X \otimes (f^* j_V) \otimes (j_X^* j_{X^*}) = \varepsilon h.$$

This also follows immediately in a strict pivotal model.

On the S-isotypic summand, h is a fixed self-duality of S times a nondegenerate bilinear form on its multiplicity space. That bilinear form has symmetry sign $\varepsilon/\nu_2(S)$. If the signs differ, it is alternating, so its dimension is even. □

The lemma uses neither a braiding nor unitarity. It does not assert preservation of a numerical indicator by an arbitrary tensor functor. In particular, it has no extra algebra-existence premise.

Application to `R8_3_1768`

Write the eight simple labels as $1, g = X_1, X_2, \ldots, X_7$. All are self-dual. The accepted complete indicator argument in the positive representative gives

$$(\nu_2(1), \nu_2(g), \nu_2(X_2), \ldots, \nu_2(X_7)) = (1, -1, 1, 1, 1, 1, 1, 1).$$

The exact tensor gives

$$X_2 g X_2^* = g + 2X_2 + 2X_3 + 2X_4 + 2X_5 + X_6 + X_7.$$

Its self-duality has negative sign by the lemma. But both positive-indicator simples X_6, X_7 occur once. Either multiplicity contradicts the lemma.

Thus the positive representative cannot exist. The inherited existence reduction takes any complex categorification to such a representative with the same tensor, giving the general-complex exclusion. Only that existence implication is used; no claim is made that an arbitrary original category already has a positive pivotal structure.

Appendix A.2. Exact complete-domain replay

The full conjugation test gives an alternative verification that does not assume the final output of the old forty-branch indicator run.

For self-dual labels Y, S, join them when some simple X has $[XYX^* : S]$ odd. Their indicators must agree by the lemma. In R1768 the graph has precisely two components:

$$\{0, 2, 3, 5\}, \qquad \{1, 4, 6, 7\}.$$

Since the unit has indicator 1, the only remaining sign vectors are

$$(1, \varepsilon, 1, 1, \varepsilon, 1, \varepsilon, \varepsilon), \qquad \varepsilon = \pm 1.$$

The exact replay tests all 128 assignments with unit sign fixed. It then reconstructs the published necessary center domains and verifies six existing Farkas certificates:

Sign choice	Complete domain	Replay
$\varepsilon = +1$	Pointed C_2 Fourier cone; 10,711 columns	Excluded
$\varepsilon = -1$	Five primitive-37 branches; 6,100 columns each	All excluded

For every separator y, the replay checks $y^T A_j \geq 0$ for *every* allowed column and $y^T b < 0$. Hence $Ax = b,\ x \geq 0$ is impossible. The negative-sign branches are indexed by

$$V = 0, 1, 2, 3, 4, \qquad U + 4V = 16.$$

The source-character, Galois-complement and five-branch arithmetic are replayed as well. Domain metadata agree with the archived systems. No optimization is invoked. The six-certificate run took about 30 seconds on this environment.

Inherited mathematical premises. The complete center-domain theorem, including the saturated conductor-37 reduction and all-order rational-ratio columns, remains essential. So does the general-complex existence reduction. Their receipt and manuscript sources are identified in the companion note `new/r1768/PALINDROMIC_EXCLUSION.md`. Exact arithmetic cannot by itself prove these categorical premises.

Source integrity. Forty published source/certificate files are pinned by SHA256 in the archive. The sole replay adapter supplies the same SHA-bound tensor in place of an obsolete whole-band census file. The equation and domain builders are unchanged.

Why it was missed. The previous parity check used the total coefficient of $\sqrt{37}$, and therefore only the sum of noninvertible multiplicities. Here that sum is ten, which is even. The new lemma tests each self-dual constituent separately and detects the two odd multiplicities. Passing the dimension-level check was insufficient.

The same full-multiplicity test was applied to the other six tensors and to the known auxiliary rank-ten R1154 candidate. None gave another contradiction.